

to speak. You could try out a Wi-Fi shield to make your project get an active network connection with the rest of the world. In today's Internet of Things driven project development scenario, some type of network connectivity is almost a minimum requirement. Of course, if you want to do more through the Internet than just maintaining network connection, you can go for a 4-relay shield that lets you control high voltage circuits by switching them on or off. Each relay can handle up to 3 amperes of current so you can hook it up to a music system, your water heater, the security system and the coffee machine. Then just use the camera module mentioned in the next section to set up a system that immediately switches on your music, makes coffee, and warms up your bathwater while deactivating the security system when you walk out of your bedroom in the morning.

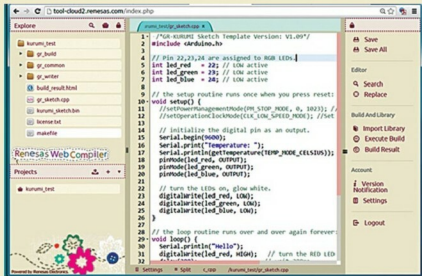
Playing With Image Processing

The comparatively more powerful MCU on the KAEDE than on the GR-SAKURA lets you implement image processing on it via a dedicated bus and on-board SDRAM. An optional CMOS camera board can be bought from Renesas, which makes it very easy to integrate the CMOS camera with your KAEDE board. The CMOS camera board allows the KAEDE to detect human beings or other things in the area where the camera is focused. It's limited to 10fps with VGA but that should suffice for most mini projects.

Peeking under the hood

As mentioned earlier, the SAKURA was based on an RX63N group MCU, while the KAEDE is based on the more powerful RX64M group MCU that lets you do much more than the older board. The newer chip also packs a core known as the RXv2, which can deliver up to 1.35 times the Coremark/MHz (4.55 Coremark/MHz) performance as well as a 40 per cent reduction in power consumption as compared to previous RX products, according to Renesas. This translates to a significant increase from the 165 Dhrystone million instructions per second (DMIPS) on the SAKURA to 240 DMIPS on the KAEDE.

The 100/144/145/176/177-pin microcontroller comes with 4/3/2.5/2 megabytes of read only memory with Trusted memory® to store your programming, and



Renesas web compiler

552 KB of random access memory (512KB for work, 32KB with ECC, 8KB for standby) for faster computing. The board operates at 3.3 volts but comes with a DC power jack rated at 5 volts. It also has a host of additional functions that are supported.

The KAEDE comes with an Ethernet port for basic Internet support, a microSD jacket, USB (Func) and a Joint Test Action Group (JTAG) interface, as well as the Arduino Shield interface that we had mentioned earlier. It can be programmed by using the USB mass storage and includes user switch, reset switch, and a user LED.

Getting Started

There are two ways to write code for this board. You can either use the web compiler that was mentioned earlier in this article, or you can try the professional integrated development environment offered by Renesas called e2studio. You can access the web compiler at the Gadget Renesas site (gadget.renesas.com) and do a guest login. The basic operations in the web compiler are explained at the RensasRulz.com website.

A new project can be created by clicking the '+' icon in the web compiler, which lets you start a project from scratch without user registration. You can then go ahead and select a template from among the template files, and click on 'create project' to get the sample code to display. To build the code, click on the "execute build" icon. You can then download the completed build, which is available to you in the form of a binary file named sketch.

All you now need to do is connect your GR-KAEDE to your computer. Once you press the reset button, the GR-KAEDE is displayed as storage, and you can copy and paste the sketch.bin file to the storage. And you're done!